Qiskit Fall Fest 2025

Tutorial 1

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1. Basic Linear Algebra for Quantum Computing

1.1 Normalize the state $|\psi\rangle = |1\rangle + 2\,|0\rangle$.

1.2 Find the value of b if $|\psi\rangle$ is normalized, where

$$|\psi\rangle = b\,|0\rangle + \frac{1+i}{\sqrt{3}}\,|1\rangle\,.$$

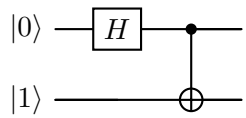
1.3 Write the outer product of $|+\rangle$ and $\langle 0|$ in matrix form.

1.4 Write down the expression of the Hadamard gate H as a linear combination of outer products of the computational basis states.

1.5 Find the eigenstates of the Hadamard gate.

1.6 Apply the H gate to the states $|0\rangle$ and $|1\rangle$, and also to $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.

1.7 Obtain the output state of the following circuit:



(1)

1.8 How many real parameters are required to describe a normalized qubit state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$? Explain why two parameters suffice.

1.9 Let $|\psi\rangle = \frac{1}{\sqrt{3}}(|0\rangle + \sqrt{2}\,|1\rangle)$. Compute the density matrix $\rho = |\psi\rangle\langle\psi|$.

1.10 Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Show that A is not unitary, and find a unitary matrix U that has the same second column as A.

1.11 Compute the tensor product

$$(\alpha\,|0\rangle + \beta\,|1\rangle) \otimes (|0\rangle - |1\rangle)$$

and express it in the computational basis.

1.12 Given the operator

$$M = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

verify that M is Hermitian and unitary, and compute its eigenvalues and eigenvectors.

2. Important Qubit Gates

2.1 What are some universal quantum gates? List a set of universal quantum gates.

2.2 Apply the Hadamard gate to $|0\rangle$ and $|1\rangle$ and verify the resulting states.

2.3 For the circuit with input $|10\rangle$ and gates H on the first qubit and X (CNOT) controlled by the first qubit on the second, determine the final two-qubit state.

2.4 Show explicitly that X, Y, and Z anti commute with each other and square to the identity.

2.5 Compute the matrix of $R_x(\theta) = e^{-i\theta X/2}$ using its power-series definition.

2.6 Let $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle)$. Compute the probability of measuring $|0\rangle$ after applying a Hadamard gate.

2.7 For $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, show that $(H \otimes I)$ followed by a CNOT produces another Bell state. Identify which one.

2.8 Determine whether

$$U = \frac{1}{2}\begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$$

is unitary, and if so, express it as a rotation about some Bloch-sphere axis.

3. Bloch Sphere Representation

3.1 Express $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle)$ on the Bloch sphere. For which ϕ does $|\psi\rangle$ lie on the X–Z plane?

3.2 For $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$, compute the expectation values $\langle X\rangle, \langle Y\rangle, \langle Z\rangle$.

3.3 Show that $R_z(\pi/2)$ acting on $|+\rangle$ gives $\frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$.

3.4 Find the Bloch vector corresponding to $\rho = \frac{1}{2}\begin{bmatrix} 1 & r \\ r^* & 1 \end{bmatrix}$. For what values of r is this a pure state?

3.5 Suppose a qubit in state $|\psi\rangle$ is measured in the basis $\{|+\rangle, |-\rangle\}$. Express the measurement probabilities in terms of its Bloch coordinates (θ, ϕ).

3.6 Can we physically distinguish between $|1\rangle$ and $-|1\rangle$? Explain why or why not.